



Methods in Calculus · Worksheet 1

Core Pure 2 Chapter 3: Methods in Calculus. Answer all questions. Show your working clearly. Total: 85 marks. Time: 105 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are four sections, one for each topic in Chapter 3. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Give exact answers where possible. For improper integrals, replace the infinite (or undefined) limit by t and show clearly the limiting process.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Improper integrals	20	
B Mean value of a function	19	
C Arc length	20	
D Surface area of revolution	21	
Roar-level challenge	5	
Total	85	

Section A Improper integrals

20 marks

1. Evaluate $\int_1^{\infty} \frac{1}{x^2} dx$.

[2]

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2. Explain why $\int_0^8 x^{-\frac{1}{3}} dx$ is an improper integral, and find its value. [3]

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3. Determine whether $\int_1^\infty \frac{1}{\sqrt{x}} dx$ converges. Give a reason for your answer. [3]

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4. Evaluate $\int_0^\infty xe^{-x} dx$. You may assume that $te^{-t} \rightarrow 0$ as $t \rightarrow \infty$. [4]

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5. (a) Show that $\int_1^{\infty} \frac{1}{x(x+1)} dx = \ln 2$.

[4]

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(b) Evaluate $\int_0^1 \ln x dx$. You may assume that $t \ln t \rightarrow 0$ as $t \rightarrow 0$.

[4]

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(Total for Question 5 is 8 marks)



Section B Mean value of a function

19 marks

6. Find the mean value of $f(x) = x^2$ over the interval $[0, 3]$. [2]

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7. Find the exact mean value of $f(x) = \sin x$ over the interval $[0, \pi]$. [3]

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8. Find the exact mean value of $f(x) = \frac{1}{1+x^2}$ over the interval $[0, 1]$. [3]

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9. Find the exact mean value of $f(x) = xe^x$ over the interval $[0, 2]$. [4]

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10. (a) Given that $y = \arctan x$, show that $\frac{dy}{dx} = \frac{1}{1+x^2}$. [2]

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(b) Find the exact mean value of $f(x) = \frac{4}{4+x^2}$ over the interval $[0, 2]$. [3]

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(c) Hence write down the mean value over the interval $[0, 2]$ of [2]

- (i) $f(x) + 3$,
- (ii) $-2f(x)$.

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(Total for Question 10 is 7 marks)



Section C Arc length

20 marks

11. The curve C has equation $y = \cosh x$. Show that $1 + \left(\frac{dy}{dx}\right)^2 = \cosh^2 x$. [2]

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12. Hence find the exact length of the curve $y = \cosh x$ between $x = 0$ and $x = \ln 2$. [3]

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13. Find the exact length of the curve $y = \frac{2}{3}x^{\frac{3}{2}}$ between $x = 0$ and $x = 3$. [3]

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14. The curve C has parametric equations [4]

$$x = t^2, \quad y = \frac{2}{3}t^3, \quad 0 \leq t \leq 2\sqrt{2}.$$

Find the exact length of C .

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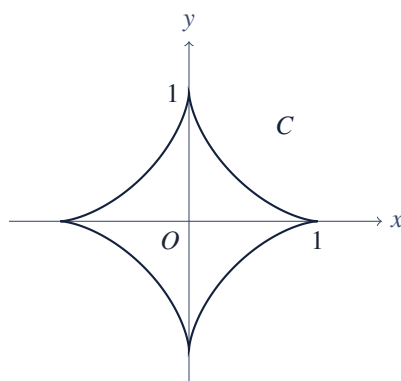
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15. The curve C , shown below, is an astroid with parametric equations

$$x = \cos^3 t, \quad y = \sin^3 t, \quad 0 \leq t < 2\pi.$$



(a) Show that $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 9\sin^2 t \cos^2 t$. [3]

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(b) Find the length of the part of C in the first quadrant. [4]

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(c) Hence write down the total length of C . [1]

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(Total for Question 15 is 8 marks)



Section D Surface area of revolution

21 marks

16. The line $y = x$, $0 \leq x \leq 2$, is rotated through 2π radians about the x -axis. Use integration to find the exact area of the curved surface formed. [2]

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17. The curve $y = 2\sqrt{x}$, $0 \leq x \leq 3$, is rotated through 2π radians about the x -axis. Find the exact area of the surface generated. [3]

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18. By rotating the curve $y = \sqrt{r^2 - x^2}$, $-r \leq x \leq r$, through 2π radians about the x -axis, show that the surface area of a sphere of radius r is $4\pi r^2$. [3]

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19. The curve $y = \cosh x$, $0 \leq x \leq \ln 2$, is rotated through 2π radians about the x -axis. Show that the area of the surface generated is $\pi \left(\ln 2 + \frac{15}{16} \right)$. [4]

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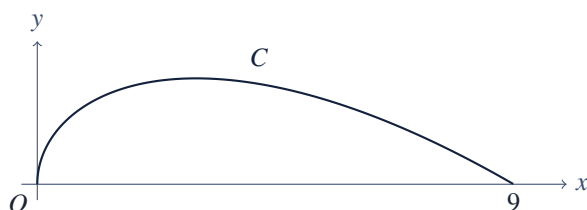
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20. The curve C has parametric equations

$$x = 3t^2, \quad y = 3t - t^3, \quad 0 \leq t \leq \sqrt{3}.$$



(a) Show that $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 9(1+t^2)^2$. [3]

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(b) Find the exact length of C . [2]

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(c) The curve C is rotated through 2π radians about the x -axis. Show that the area of the surface generated is 27π . [4]

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(Total for Question 20 is 9 marks)



ROAR-LEVEL CHALLENGE

5 marks

21. Show that $\int_{-\infty}^{\infty} \frac{1}{x^2 + 2x + 5} dx$ converges, and find its exact value.

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END OF WORKSHEET

Total: 85 marks